

PRACTICE CHALLENGE

# SMART ESCAPE

## Interactive Evacuation Route Simulator

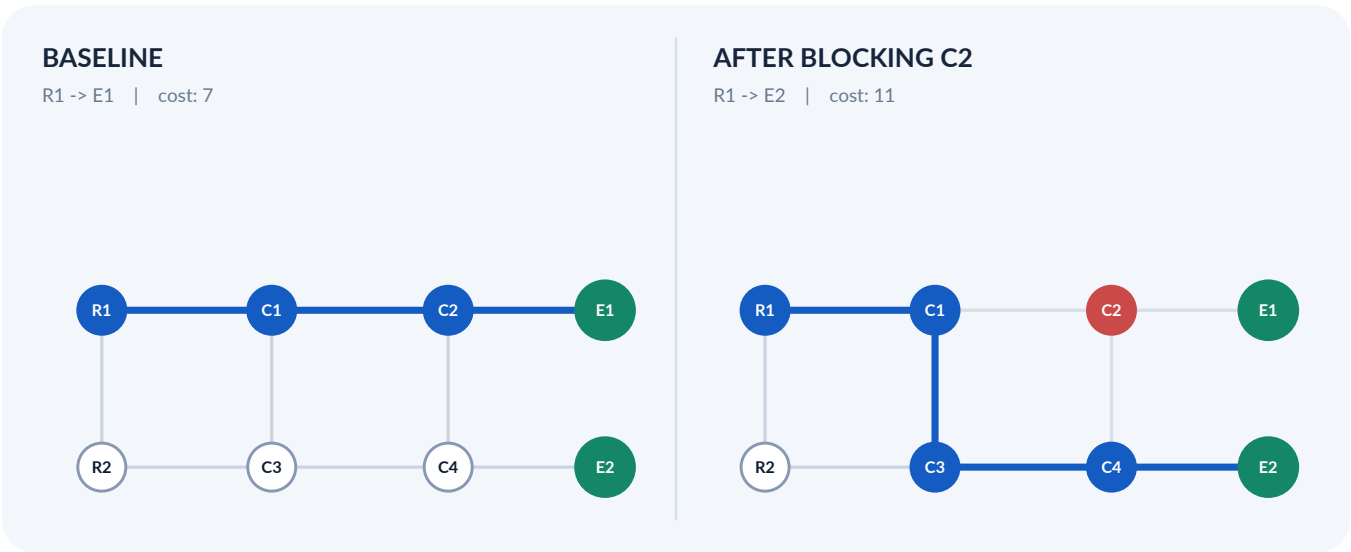
BUILD AN INTERACTIVE MAP. COMPUTE ROUTES. RESPOND TO CHANGING HAZARDS.

### 01 Problem

Imagine a building where a corridor is obstructed, a room becomes unavailable, or an exit is closed. These are **simulated hazards**. Build **Smart Escape**, a browser-based tool that displays the building map and finds the lowest-cost route from a chosen starting location to an open exit. When hazards change, the application must immediately identify an alternative route or clearly report that none exists.

| FORMAT         | TIME LIMIT | APPLICATION           |
|----------------|------------|-----------------------|
| Solo mock test | 90 minutes | Frontend-only web app |

### THE CENTRAL INTERACTION



Baseline and rerouting shown using the supplied sample. Diagram lengths do not determine path costs.

### 02 Provided data

At T+0, you receive **building.json**. Import it locally in the browser; the judges will also use unseen datasets with the same schema. The graph describes rooms, junctions, exits, weighted corridors and initial hazard states.

**Important:** Smart Escape is an educational simulation, not a certified real-world evacuation planning tool. The current **AI DevFest Official Rulebook** also applies.

## 03 Input and functional requirements

### 3.1 Input format

Read the provided JSON file. All fields below are required. Empty initial-state arrays are valid.

| FIELD         | DEFINITION                                                                                                                                              |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| building      | Non-empty building name.                                                                                                                                |
| nodes[]       | Each node has a unique <b>id</b> , non-empty <b>label</b> , <b>type</b> = room   junction   exit, and numeric display coordinates <b>x</b> , <b>y</b> . |
| edges[]       | Each edge has a unique <b>id</b> , valid <b>from</b> and <b>to</b> node IDs, and a positive integer <b>cost</b> .                                       |
| initial_state | Arrays <b>blocked_nodes</b> , <b>blocked_edges</b> , <b>closed_exits</b> , containing IDs of currently unavailable elements.                            |

**Input limits:** 2-60 nodes and 1-150 undirected edges; at least one room/junction and one exit. IDs are case-sensitive. No self-loops or repeated node pairs. Initial-state IDs must exist and match their category (blocked nodes are rooms/junctions; closed exits are exits). Disconnected graphs are valid. Reject malformed or inconsistent files clearly.

### 3.2 Mandatory tasks

- **Import and map:** Validate input and display all nodes and edges at the supplied coordinates with readable labels, distinct node types and visible corridor costs.
- **Select and calculate:** Let the user select an unblocked room or junction as the start. Highlight the lowest-cost route to an accessible exit, showing its node sequence, exit and total cost.
- **Change conditions:** Allow users to block/unblock rooms or junctions and corridors, and close/reopen exits. Distinguish these states visually.
- **Update and reset:** Recalculate immediately after every start or hazard change, without reimporting. Reset must restore the file's original **initial\_state**.
- **Handle failure cases:** Show **No route available** when no exit is reachable and **Starting location blocked** if the selected start becomes blocked.
- **Two languages:** Provide Bangla and English modes for all principal labels, buttons, statuses, errors and instructions. Dataset labels may remain unchanged.

### 3.3 Exact routing rules

- Calculate route cost as the **sum of edge costs**. Display coordinates and number of corridors must not be used as a substitute for cost.
- Exclude blocked nodes and their incident edges, blocked edges, and closed exits (including as intermediate nodes).
- Choose the reachable open exit with minimum cost. On equal cost, choose the lexicographically smallest **exit ID**; if paths to that exit also tie, choose the lexicographically smallest sequence of node IDs.

### 3.4 Understanding simulated hazards

- A **blocked room or junction** cannot be entered or crossed, and all corridors attached to it become unusable. A **blocked corridor** removes only that connection; its end locations can still be reached by other routes.
- A **closed exit** cannot be used as a destination. For example, blocking **C2** in the sample removes the original route to E1. The application must show the new least-cost route through C3 and C4 to E2.
- Hazards are **manually changed by the user**; detecting fires or predicting hazard spread is not required. Every change must update the route, while **Reset** restores the supplied initial state.

## 04 Test cases and submission

### 4.1 Sample checks

Using the provided `building.json`, each test below starts from `initial_state` unless stated otherwise.

| SCENARIO         | ACTION                     | EXPECTED RESULT                        |
|------------------|----------------------------|----------------------------------------|
| Baseline         | Select R1                  | R1 - C1 - C2 - E1; cost <b>7</b>       |
| Blocked junction | Select R1; block C2        | R1 - C1 - C3 - C4 - E2; cost <b>11</b> |
| Exits closed     | Select R1; close E1 and E2 | <b>No route available</b>              |
| Different start  | Select R2                  | R2 - C3 - C4 - E2; cost <b>7</b>       |
| Blocked start    | Select R1; then block R1   | <b>Starting location blocked</b>       |

Judges may test unseen graphs, equal-cost ties, disconnected areas and invalid input. Hard-coded sample routes are not acceptable.

### 4.2 Visual animation and optional extensions

**Required:** Use subtle animations for selecting locations, toggling hazards and updating highlighted routes. Keep transitions brief and readable; avoid flashing or delaying controls. **Optional:** Alternative routes, accessible/high-contrast controls, PNG export, saving progress and advanced route walkthroughs.

## 05 Essential competition rules

- **Start and duration:** Work individually for **90 minutes**, beginning at T+0. Create a new public repository named `devfest-<registration-number>`; no pre-existing project code or code committed before T+0. Permitted libraries and official starter tools are allowed.
- **Git commits:** Commit **at least once every 30 minutes**, with a **minimum of three commits** in total. Each message must state what changed and include the AI prompt used, or **Manual edit**. Do not rewrite Git history.
- **Technical restrictions:** Frontend only; no participant-controlled backend, serverless functions, or persistent remote databases/storage. Browser storage and Rulebook-permitted HTTPS/CORS APIs are allowed; routing must work without external APIs. Optional in-app AI uses a user-supplied key. Do not include secrets.
- **Deadline:** Final commit must be **created and pushed by T+90**. The matching public HTTPS deployment is **required by T+90** and must work in the latest Chrome without login or installation. Stop code, Git and deployment changes at T+90.

## 06 Deliverables (required)

Submit through the **official submission form** by T+90:

- **Full name and registration number**, plus your **public GitHub repository URL**.
- **Final commit ID:** the complete hash or its first seven characters.
- **Public HTTPS live website link (mandatory)**, hosted on GitHub Pages, Vercel, Netlify, Cloudflare Pages or equivalent; accessible to judges **without login or special permission**.

Your repository must include **all source code**, **README.md** (identity, live link, running instructions, implemented/bonus features, known issues, AI tools and most useful prompt), a **LICENSE** file containing the MIT License, and **screenshots/** showing both the baseline route and rerouting after C2 is blocked.

All other provisions of the current **Official Rulebook**, including late-submission conditions, remain in force.